Chapter 3: Expectation of Random Variables

AMS 310
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(1) Mean and Variance of Discrete R.V.s

Definition: Mean (Expected value) of a discrete r.v. X:

$$E(X) = \mu = \sum_{\text{all } x} xp(x)$$

Example 3.1: What is the expected number of heads in three tosses of a fair coin?

X: # heads in three tosses of a fair coin

x	$p(x)$	$xp(x)$
0	$\frac{1}{8}$	0
1	$\frac{3}{8}$	$\frac{3}{8}$
2	$\frac{3}{8}$	$\frac{3}{4}$
3	$\frac{1}{8}$	$\frac{3}{8}$
Total	1	$\mu = 1.5$

$$E\big(h(X)\big) = \sum_{\text{all } x} h(x)p(x)$$

<u>Example 3.2</u>: In flipping 3 balanced coins find $E(X^3 - X)$

x	$x^3 - x$	$p(x)$	$(x^3 - x)p(x)$
0	0	1/8	0
1	0	3/8	0
2	6	3/8	9/4
3	24	1/8	3
Total		1	21/4

$$E(X^3 - X) = 0 + 0 + \frac{9}{4} + 3 = \frac{21}{4} = 5.25$$

<u>Variance</u> of a discrete r.v. X:

$$Var(X) = \sigma^2 = E[(X-\mu)^2] = \sum_{\text{all } x} (x-\mu)^2 p(x)$$

$$\text{sd}(X) = \sigma = \sqrt{Var(X)}$$

Alternatively $\quad Var(X) = E(X^2) - \mu^2 = \sum_{\text{all } x} x^2 p(x) - \left[\sum_{\text{all } x} x p(x)\right]^2$

Proof:

$$Var(X) = E(X-\mu)^2 = \sum_{\text{all } x} (x-\mu)^2 p(x)$$

$$= \sum_{\text{all } x} (x^2 - 2\mu x + \mu^2) p(x)$$

$$= \sum_{\text{all } x} x^2 p(x) - 2\mu \sum_{\text{all } x} x p(x) + \mu^2 \sum_{\text{all } x} p(x)$$

$$= \sum_{\text{all } x} x^2 p(x) - 2\,\mu^2 + \mu^2 = \sum_{\text{all } x} x^2 p(x) - \mu^2 = E(X^2) - \mu^2$$

Example 3.3: Mean and variance

x	$p(x)$	$xp(x)$	$(x-\mu)^2$	$(x-\mu)^2 p(x)$
1	0.3	0.3	4	1.2
2	0.4	0.8	1	0.4
5	0.2	1.0	4	0.8
9	0.1	0.9	36	3.6
Total	1	$\mu = 3$		$\sigma^2 = 6$

Example 3.4: Alternative way to calculate the variance

x	x^2	$p(x)$	$xp(x)$	$x^2 p(x)$
1	1	0.3	0.3	0.3
2	4	0.4	0.8	1.6
5	25	0.2	1.0	5.0
9	81	0.1	0.9	8.1
Total		1	$\mu = 3$	$E(X^2) = 15$

$$Var(X) = E(X^2) - \mu^2 = 15 - 3^2 = 6$$

Bernoulli RV's

Let $X \sim \text{Bernoulli}(p)$. Then

$$p(x) = p^x(1-p)^{1-x}, \qquad x = 0, 1$$

$$E(X) = \sum_{x=0}^{1} x p^x (1-p)^{1-x} = p$$

$$E(X^2) = \sum_{x=0}^{1} x^2 p^x (1-p)^{1-x} = p$$

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = p - p^2 = p(1-p)$$

Binomial RV's

$X = \sum_{i=1}^{n} Y_i \sim \mathrm{Bin}(n, p)$, where $Y_1, Y_2, \cdots, Y_n$ are independent Bernoulli(p).

$$E(X) = E\left(\sum_{i=1}^{n} Y_i\right) = \sum_{i=1}^{n} E(Y_i) = \sum_{i=1}^{n} p = np$$

$$\mathrm{Var}(X) = \mathrm{Var}\left(\sum_{i=1}^{n} Y_i\right) = \sum_{i=1}^{n} \mathrm{Var}(Y_i)$$

$$= \sum_{i=1}^{n} p(1-p) = np(1-p)$$

Binomial Distribution

For $X \sim \text{Bin}(n, p)$,
Expected value $= E(X) = \mu = np$
Variance $= Var(X) = \sigma^2 = np(1-p)$

<u>Example 3.5</u>: Find the mean and variance of the probability distribution of the number of heads obtained in three flips of a balanced coin.

$$X \sim \text{Bin}(3, 0.5)$$

$$E(X) = \mu = np = 3(0.5) = 1.5$$

$$Var(X) = \sigma^2 = np(1-p) = 3(0.5)(0.5) = 0.75$$

Hypergeometric RV

- Sampling w/o replacement

1) Finite population with N individuals (binomial: infinite population or with replacement)
2) S or F. There are m successes in the population.
3) Sample size: n. Each subset of size n is equally likely to be chosen.

$$P(X = x) = \frac{\binom{m}{x}\binom{N-m}{n-x}}{\binom{N}{n}}$$

$$\max\{0, n - N + m\} \leq x \leq \min\{n, m\}$$

$$(\because x \leq m \text{ and } n - x \leq N - m)$$

$$E(X) = \frac{nm}{N}, \qquad Var(X) = \frac{nm(N-m)(N-n)}{N^2(N-1)}$$

$$E(X) = \frac{nm}{N}$$

$X = \sum_{i=1}^{n} X_i$, where X_i is the ith item selected is success.

(a)

$$E(X) = E\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} E\left(X_i\right)$$

$$= \sum_{i=1}^{n} \frac{m}{N} = \frac{nm}{N}$$

(b)$\mathrm{Var}(X) = \sum_{i=1}^{n} \mathrm{Var}(X_i) + 2\sum_{i<j}\sum \mathrm{Cov}\left(X_i, X_j\right)$

$$\mathrm{Var}(X_i) = p(1-p) = \frac{m}{N}\left(1 - \frac{m}{N}\right)$$

$$\mathrm{Cov}\left(X_i, X_j\right) = \mathrm{E}\left(X_i X_j\right) - \mathrm{E}(X_i)\mathrm{E}\left(X_j\right)$$

$$\mathrm{E}\left(X_i X_j\right) = \mathrm{P}\left(X_i X_j = 1\right) = \mathrm{P}\left(X_i = 1, X_j = 1\right)$$

$$= \mathrm{P}(X_i = 1)\mathrm{P}\left(X_j = 1 \mid X_i = 1\right) = \frac{m}{N}\left(\frac{m-1}{N-1}\right)$$

$$\mathrm{Var}(X) = n \cdot \frac{m}{N}\left(1 - \frac{m}{N}\right) + n(n-1)\left[\frac{m}{N}\left(\frac{m-1}{N-1}\right) - \left(\frac{m}{N}\right)^2\right]$$

$$= n \cdot \frac{m}{N}\left[\left(1 - \frac{m}{N}\right) + (n-1)\left(\frac{m-1}{N-1} - \frac{m}{N}\right)\right]$$

$$= n \cdot \frac{m}{N}\left[\frac{N-m}{N} + \frac{(n-1)\{N(m-1) - m(N-1)\}}{N(N-1)}\right]$$

$$= n \cdot \frac{m}{N}\left[\frac{N-m}{N} - \frac{(n-1)(N-m)}{N(N-1)}\right] = n \cdot \frac{m}{N} \cdot \frac{N-m}{N}\left(1 - \frac{n-1}{N-1}\right)$$

$$= n \cdot \frac{m}{N}\left(1 - \frac{m}{N}\right)\frac{N-n}{N-1}$$

Hypergeometric RV

$$E(X) = \frac{nm}{N}, \qquad \mathrm{Var}(X) = \frac{nm(N-m)(N-n)}{N^2(N-1)}$$

Let $\frac{m}{N} = p$. Then

$E(X) = np$, $\mathrm{Var}(X) = np(1-p)\left(\frac{N-n}{N-1}\right)$

If $N \gg n$, then $\mathrm{Var}(X) \approx np(1-p)$

$\frac{N-n}{N-1}$: finite population correction factor

Binomial approximation to the hypergeometric, when

$\frac{n}{N} \leq 0.1$, and $p = \frac{m}{N}$ is not too close to either 0 or 1.

Example 3.6

For a lot of 100 CD's, 20 are defective. Find the probability that among a randomly selected sample of 10 CD's, 2 are defective, by using

(a) The formula for the hypergeometric distribution:

$$x = 2, n = 10, m = 20, N = 100$$

$$p(2) = \frac{\binom{20}{2}\binom{80}{8}}{\binom{100}{10}} = 0.318$$

(b) The binomial approximation to the hypergeometric:

$$x = 2, n = 10, p = \frac{a}{N} = \frac{20}{100} = 0.2, \frac{n}{N} = 0.1$$

$$p(x) = \binom{10}{2}(0.2)^2(0.8)^8 = 0.302 \text{ (close)}$$

$X \sim \text{Poisson}(\lambda)$

$$E(X) = \sum_{x=0}^{\infty} x \frac{\lambda^x e^{-\lambda}}{x!} = \sum_{x=1}^{\infty} \frac{\lambda^x e^{-\lambda}}{(x-1)!} = \lambda \sum_{k=0}^{\infty} \frac{\lambda^k e^{-\lambda}}{k!} = \lambda$$

$$E(X^2) = \sum_{x=0}^{\infty} x^2 \frac{\lambda^x e^{-\lambda}}{x!} = \sum_{x=0}^{\infty} \frac{[x(x-1)+x]\lambda^x e^{-\lambda}}{x!}$$

$$= \sum_{x=0}^{\infty} \frac{x(x-1)\lambda^x e^{-\lambda}}{x!} + \sum_{x=0}^{\infty} \frac{x\lambda^x e^{-\lambda}}{x!} = \lambda^2 \sum_{x=2}^{\infty} \frac{\lambda^{x-2} e^{-\lambda}}{(x-2)!} + E(X)$$

$$= \lambda^2 \sum_{k=0}^{\infty} \frac{\lambda^k e^{-\lambda}}{k!} + \lambda = \lambda^2 + \lambda$$

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = \lambda^2 + \lambda - \lambda^2 = \lambda$$

Poisson Approximation to the Binomial

Let $X \sim \mathrm{Bin}(n, p)$.

Let $n \to \infty$ and $p \to 0$ such that np remains constant (np is a moderate number). Then

X is approximately $\mathrm{Poisson}(\lambda)$ such that $\lambda = np$.

$n \geq 20, p \leq 0.05$: acceptable

$n \geq 100, p \leq 0.01$: excellent approximation

$X \sim \mathrm{Bin}(n, p) \Longrightarrow E(X) = np = \lambda$

$Var(X) = np(1-p) = \lambda(1-p) \to \lambda$ as $n \to \infty$ & $p \to 0$

Poisson Approximation to the Binomial

<u>Alternative Approach</u>

$X \sim \mathrm{Bin}(n, p), \lambda = np, n \to \infty$

$$\Longrightarrow P(X = x) = \frac{n!}{(n-x)!\,x!} p^x (1-p)^{n-x} = \frac{n!}{(n-x)!\,x!} \left(\frac{\lambda}{n}\right)^x \left(1 - \frac{\lambda}{n}\right)^{n-x}$$

$$= \frac{n(n-1)\cdots(n-x+1)}{n^x} \frac{\lambda^x}{x!} \frac{[(1-(\lambda/n)]^n}{[(1-(\lambda/n)]^x}, \qquad x = 0, 1, 2, \cdots$$

For $n \to \infty$,

$\left(1 - \frac{\lambda}{n}\right)^n \approx e^{-\lambda}$, $\frac{n(n-1)\cdots(n-x+1)}{n^x} \approx 1$, $\left(1 - \frac{\lambda}{n}\right)^n \approx 1$

Hence,

$$P(X = x) \approx \frac{\lambda^x e^{-\lambda}}{x!}, \; x = 0, 1, 2, \cdots$$

Poisson Approximation to the Binomial

$X \sim \text{Poisson}(\lambda)$

$$\Longrightarrow P(X = x) = \frac{\lambda^x e^{-\lambda}}{x!} = \frac{\lambda}{x} P(X = x - 1), \qquad x = 1, 2, \cdots$$

$Y \sim \text{Bin}(n, p)$

$$\Longrightarrow P(Y = y) = \binom{n}{y} p^y (1-p)^{n-y} = \frac{n - y - 1}{y} \cdot \frac{p}{1-p} P(Y = y - 1)$$

$$= \frac{np - p(y-1)}{y - yp} P(Y = y - 1) \approx \frac{\lambda}{y} P(Y = y - 1), \qquad y = 1, 2, \cdots, n$$

If $\lambda = np$ and $p \approx 0$,

$$P(Y = 0) = (1-p)^n = \left(1 - \frac{np}{n}\right)^n = \left(1 - \frac{\lambda}{n}\right)^n \to e^{-\lambda} = P(X = 0)$$

as $n \to \infty$

<u>Example 3.7</u>: A publisher of mystery novels tries to keep its books free of typos. The probability of any given page containing at least one typo is 0.003 and errors are independent from page to page. What is the approximate probability that a 500 page book has

(a) exactly 1 page with typos?

If X is the number of pages with typos, then $X \sim \mathrm{Bin}(n, p)$.

$$n = 500, p = 0.003 \Longrightarrow \lambda = np = (500)(0.003) = 3/2$$

$$P(X = 1) \approx \frac{(3/2)e^{-3/2}}{1!} = 0.3347$$

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>dpois(1, 1.5)
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(b) at most 2 pages with typos?

$$P(X \leq 2) \approx \sum_{x=0}^{2} \frac{1.5^x e^{-1.5}}{x!} = \left(1 + \frac{3}{2} + \frac{(3/2)^2}{2}\right) e^{-1.5} = \frac{29}{8} e^{-1.5} = 0.809$$

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>ppois(2, 1.5)
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Geometric Distribution

1. Each trial: Success (S) or Failure (F)
2. Independent trials
3. $P(S) = p$
4. Continues until the first success is observed

- Probability of getting the first success on the x-th trial:

$$f(x) = p(1-p)^{x-1}, x = 1, 2, \cdots$$
$$E(X) = \frac{1}{p}$$

- Using R,
  probability distribution
  >dgeom(x-1, p)
  cdf
  >pgeom(x-1, p)

Geometric RV

$$E(X) = \frac{1}{p}$$

<u>Proof</u>

$$E(X) = \sum_{x=1}^{\infty} xp(x) = \sum_{x=1}^{\infty} xp(1-p)^{x-1}$$

$$(1-p)E(X) = \sum_{x=1}^{\infty} xp(1-p)^{x} = \sum_{k=2}^{\infty} (k-1)p(1-p)^{k-1}$$

$$= \sum_{x=2}^{\infty} (x-1)p(1-p)^{x-1}$$

$$pE(X) = E(X) - (1-p)E(X) = \sum_{x=1}^{\infty} p(1-p)^{x-1} = \frac{p}{p} = 1$$

$$\therefore E(X) = \frac{1}{p}$$

Geometric RV

$$E(X) = \frac{1}{p}$$

<u>Altertative proof</u>

Let $q = 1 - p$. Then

$$p(x) = p(1-p)^{x-1} = (1-q)q^{x-1}, x = 1, 2, \cdots$$

$$E(X) = \sum_{x=1}^{\infty} xp(x) = \sum_{x=1}^{\infty} xp(1-p)^{x-1} = (1-q)\sum_{x=1}^{\infty} xq^{x-1}$$

Define $h(q) = \sum_{x=1}^{\infty} q^x = \frac{q}{1-q}$. Then

$$h'(q) = \sum_{x=1}^{\infty} xq^{x-1} = \frac{1}{(1-q)^2}$$

$$\therefore E(X) = (1-q)h'(q) = \frac{1}{1-q} = \frac{1}{p}$$

Geometric RV

$$Var(X) = \frac{1-p}{p^2}$$

<u>Proof</u>

Let $q = 1 - p$. Then

$$E(X^2) = \sum_{x=1}^{\infty} x^2 p(x) = \sum_{x=1}^{\infty} x^2 p(1-p)^{x-1} = (1-q)\sum_{x=1}^{\infty} x^2 q^{x-1}$$

Define $h(q) = \sum_{x=1}^{\infty} x q^x = \frac{q}{1-q}(1-q)\sum_{x=1}^{\infty} x q^{x-1} = \frac{q}{1-q}E(X) = \frac{q}{(1-q)^2}$. Then

$$h'(q) = \frac{(1-q)^2 + 2q(1-q)}{(1-q)^4} = \frac{1+q}{(1-q)^3}$$

$$\frac{1+q}{(1-q)^2} = (1-q)h'(q) = (1-q)\frac{d}{dq}\sum_{x=1}^{\infty} x q^x = (1-q)\sum_{x=1}^{\infty} x^2 q^{x-1} = E(X^2)$$

$$\therefore Var(X) = E(X^2) - [E(X)]^2 = \frac{1+q}{(1-q)^2} - \frac{1}{p^2} = \frac{2-p}{p^2} - \frac{1}{p^2} = \frac{1-p}{p^2}$$

Negative Binomial RV: $X \sim \mathrm{NB}(r, p)$

- Independent Bernoulli Trials
- X: number of trials until r-th success
- $r = 1 \Longrightarrow X \sim$Geometric$(p)$

$$p(x) = \binom{x-1}{r-1} p^r (1-p)^{x-r}, x = r, r+1, \cdots$$

$$E(X) = \frac{r}{p}$$

$$\mathrm{Var}(X) = \frac{r(1-p)}{p^2}$$

Negative Binomial RV

$$E(X) = \frac{r}{p}$$

<u>Proof</u>

$$E(X) = \sum_{x=r}^{\infty} x \binom{x-1}{r-1} p^r (1-p)^{x-r}$$

$$= \sum_{x=r}^{\infty} x \frac{(x-1)!}{(r-1)!\,(x-r)!} p^r (1-p)^{x-r} = \frac{r}{p} \sum_{x=r}^{\infty} \frac{x!}{r!\,(x-r)!} p^{r+1} (1-p)^{x-r}$$

$$= \frac{r}{p} \sum_{x=r}^{\infty} \binom{x}{r} p^{r+1} (1-p)^{x-r} = \frac{r}{p} \sum_{y=k}^{\infty} \binom{y-1}{k-1} p^k (1-p)^{y-k} = \frac{r}{p}$$

by setting $y = x + 1$ and $k = r + 1$

$$\therefore E(X) = \frac{r}{p}$$

Negative Binomial RV

$$\mathrm{Var}(X) = \frac{r(1-p)}{p^2}$$

<u>Proof</u>

$$E(X^2) = \sum_{x=r}^{\infty} [x(x+1) - x] \binom{x-1}{r-1} p^r (1-p)^{x-r}$$

$$= \sum_{x=r}^{\infty} \frac{(x+1)!}{(r-1)!\,(x-r)!} p^r (1-p)^{x-r} - E(X)$$

$$= \frac{r(r+1)}{p^2} \sum_{x=r}^{\infty} \binom{x+1}{r+1} p^{r+2} (1-p)^{x-r} - \frac{r}{p}$$

$$= \frac{r(r+1)}{p^2} \sum_{y=k}^{\infty} \binom{y-1}{k-1} p^k (1-p)^{y-k} = \frac{r(r+1)}{p^2} - \frac{r}{p}$$

by setting $y = x + 2$ and $k = r + 2$

$$\therefore \mathrm{Var}(X) = E(X^2) - [E(X)]^2 = \frac{r(r+1)}{p^2} - \frac{r}{p} - \left(\frac{r}{p}\right)^2 = \frac{r(1-p)}{p^2}$$

Negative Binomial RV

- Let $X_1, X_2, \cdots, X_n$ be independent with
$$X_i \sim \text{Geometric}(p), i = 1, 2, \cdots, n.$$

Then

$$Y = \sum_{i=1}^{n} X_i \sim \text{NB}(n, p).$$

- Let $X_1, X_2, \cdots, X_n$ be independent with
$$X_i \sim \text{NB}(r, p), i = 1, 2, \cdots, n.$$

Then

$$Y = \sum_{i=1}^{n} X_i \sim \text{NB}(rn, p).$$

Example 3.8

A pipe-smoking mathematician carries, at all times, 2 matchboxes, 1 in his left-hand pocket and 1 in his right-hand pocket. Each time he needs a match his is equally likely to take it from either pocket. Consider the moment when the mathematician first discovers that one of his matchboxes is empty. If it is assumed that both matchboxes initially contained N matches, what is the probability that there are exactly k matches in the other box, $k = 0, 1, \cdots, N$?

Answer to Example 3.8

Let E denote the event that the mathematician first discovers that the right-hand matchbox is empty and there are k matches in the left-hand box at the time. Now, this event will occur if and only if the $(N+1)$st choice of the right-hand matchbox is made at the $(N+1+N-k)$th trial. Hence, with $p=\frac{1}{2}, r=N+1$, and $x=2N-k+1$, we see

$$P(E)=\binom{2N-k}{N}\left(\frac{1}{2}\right)^{2N-k+1}$$

As there is an equal probability that it is the left-hand box that is first discovered to be empty and there are k matches in the right-hand box at that time, the desired result is

$$2P(E)=\binom{2N-k}{N}\left(\frac{1}{2}\right)^{2N-k}$$

Example 3.9

Find the expected value and the variance of the number of times one must throw a die until the outcome 1 has occurred 4 times.

Since the random variable of interest is NB$(4, 1/6)$, we get

$$E(X) = \frac{r}{p} = 24$$

$$Var(X) = \frac{r(1-p)}{p^2} = \frac{4(5/6)}{(1/6)^2} = 120$$